

NAME: ADOYI NATHANIEL JOHNSON

STUDENT ID: FWSD2511203

STUDENT EMAIL: fwsd2511203@students.icdfa.edu.ng

COURSE CODE: SBT-DF203-BASIC NETWORKING SKILLS

LAB NAME: LAB6 FIREWALL TRAFFIC CONTROL AND FORENSICS VERIFICATION

DATE: 18/09/26

INSTRUCTOR: AMINU IDRIS

International Cybersecurity and Digital Forensics Academy (ICDFA)

SBT-DF203: Basic Networking Skills for Digital Forensics

LAB 6: Firewall Traffic Control and Forensic Verification

Three-Week Delivery Block 2/3 | 12-18 September 2026

Prepared by: Aminu Idris, AMCPN | Founder, ICDFA

Important Pre-Lab Instruction

Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.

Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 6
Lab Title	Firewall Traffic Control and Forensic Verification
Delivery Block	2/3 of 3
Scheduled Dates	12-18 September 2026
Estimated Duration	4-5 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	Baseline and blocked HTTP PCAPNG files; iptables ruleset exports

1. Source Materials and Slide Alignment

- Firewall: network-based and host-based firewalls, iptables chains and rules, ACCEPT/DROP actions, UFW relationship, NFQUEUE concept and HTTP blocking.

- The slide practical lists rules, adds/removes an INPUT rule, uses a local Apache page and observes allowed versus blocked hosts.
- The homework requires traffic generated by curl from a different machine and forensic observation of the result.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

You are validating whether a Linux host-based firewall correctly blocks HTTP traffic from one authorized lab client while allowing another. You will capture baseline traffic, apply a narrowly scoped iptables rule, capture the blocked connection behaviour, prove the rule hit using counters, remove the rule and document the evidential differences.

3. Learning Outcomes

- Explain the purpose of host-based and network-based firewalls.
- Interpret INPUT, OUTPUT and FORWARD chains.
- Create, verify and remove a narrowly scoped iptables rule.
- Capture allowed and blocked HTTP connection attempts.
- Correlate packet evidence with firewall rule counters and application behaviour.
- Explain DROP versus REJECT and identify retransmission/timeout evidence.
- Restore the original ruleset safely.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.
- Before making changes, export the existing ruleset. Apply only a source-specific rule for the assigned lab IP. Never flush all firewall rules on a remotely administered or production system.

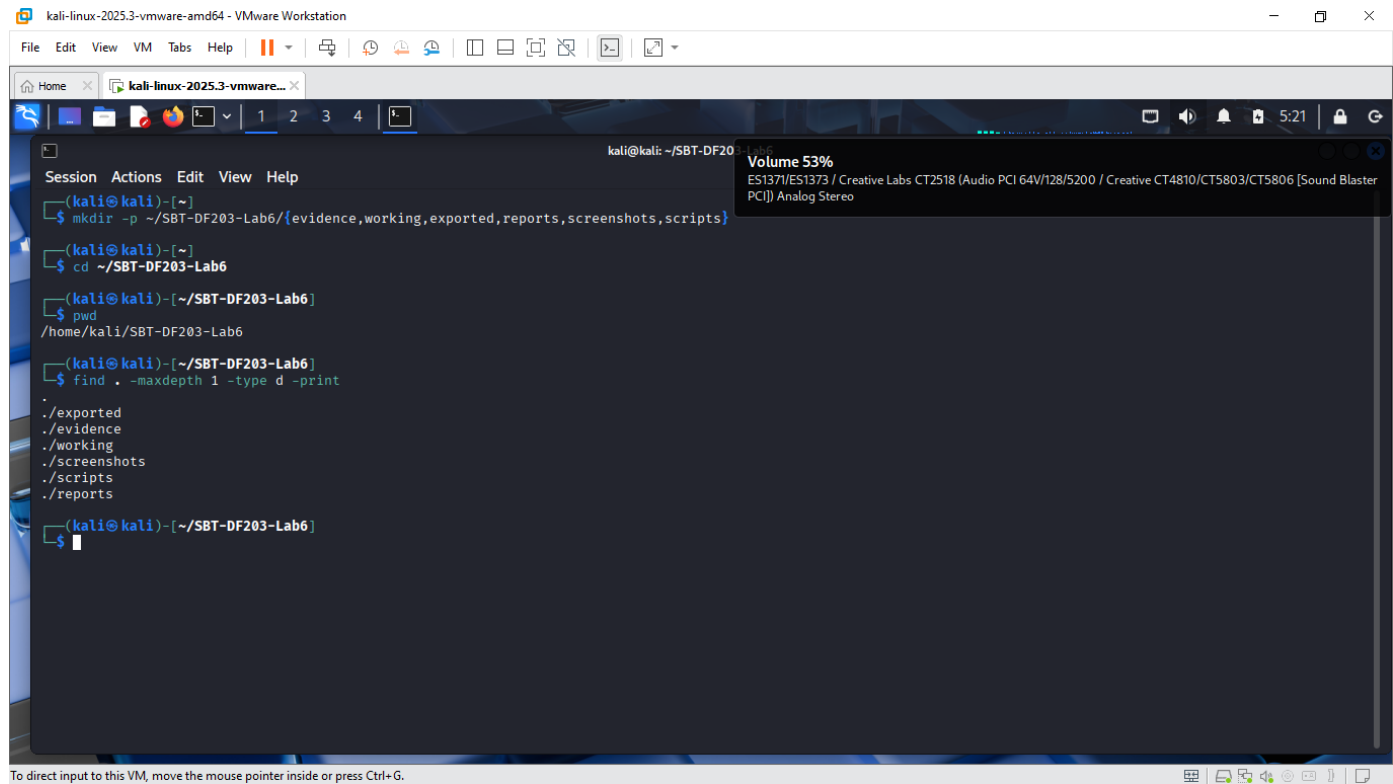
5. Required Tools and Environment

Item	Recommended Tool	Purpose
Server VM	Kali/Ubuntu with Apache	Protected host.
Client VM	Linux or Windows on host-only network	Generate HTTP requests.
Firewall tools	iptables, iptables-save	Apply and preserve rules.
Capture tools	tshark/Wireshark	Capture baseline and blocked traffic.
Client tools	curl, browser	Test access.
Optional extension	NetfilterQueue	Instructor-led packet interception demonstration.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab6/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab6
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

ANSWER:

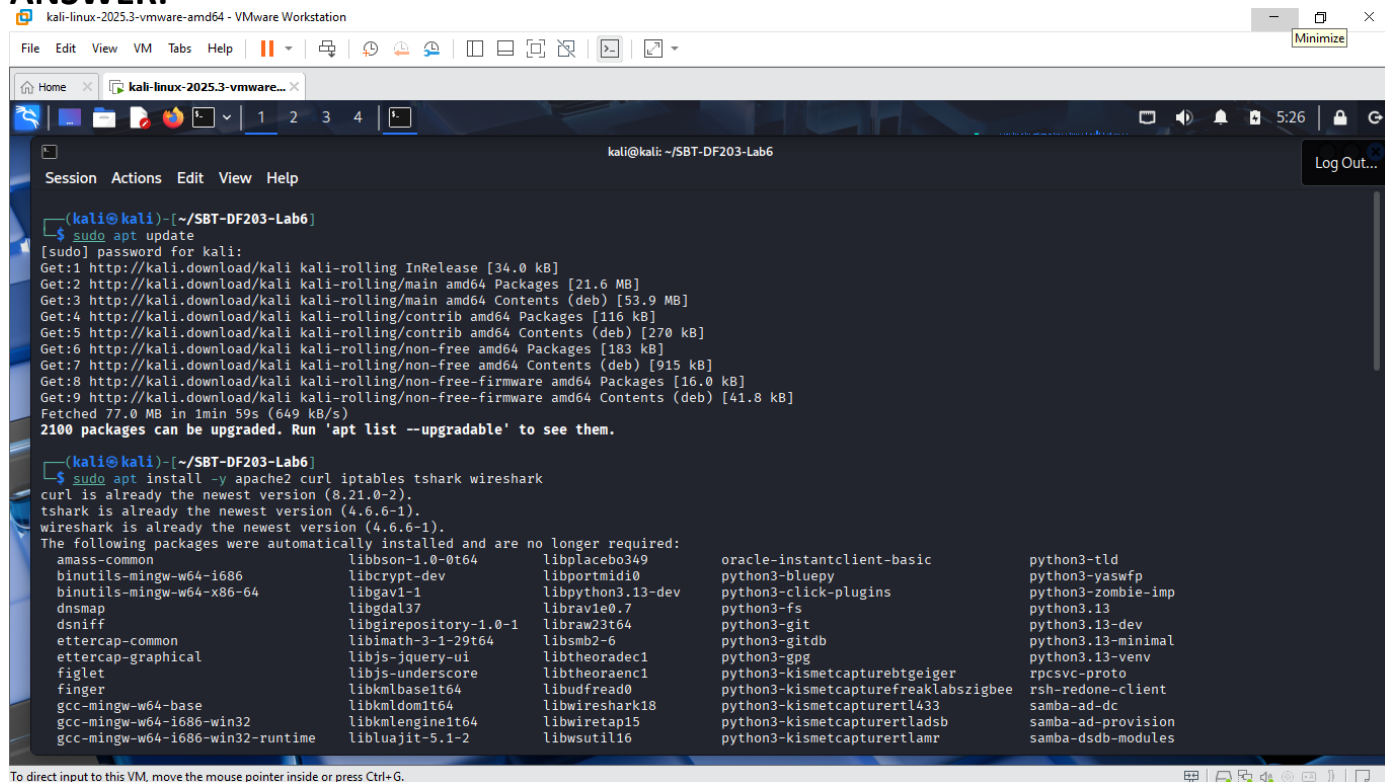
```

sudo apt update
sudo apt install -y apache2 curl iptables tshark wireshark
sudo systemctl enable --now apache2
printf '<!DOCTYPE html>\n<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: YOUR SERVER NAME</p></body></html>\n' | sudo tee /var/www/html/firewall_lab.html
sudo iptables-save | tee reports/iptables_before.rules
sudo iptables -L -n -v --line-numbers | tee reports/iptables_before.txt
sha256sum reports/iptables_before.rules | tee reports/iptables_before_sha256.txt

```

Screenshot required: Original firewall ruleset, rule listing, hash and the training webpage.

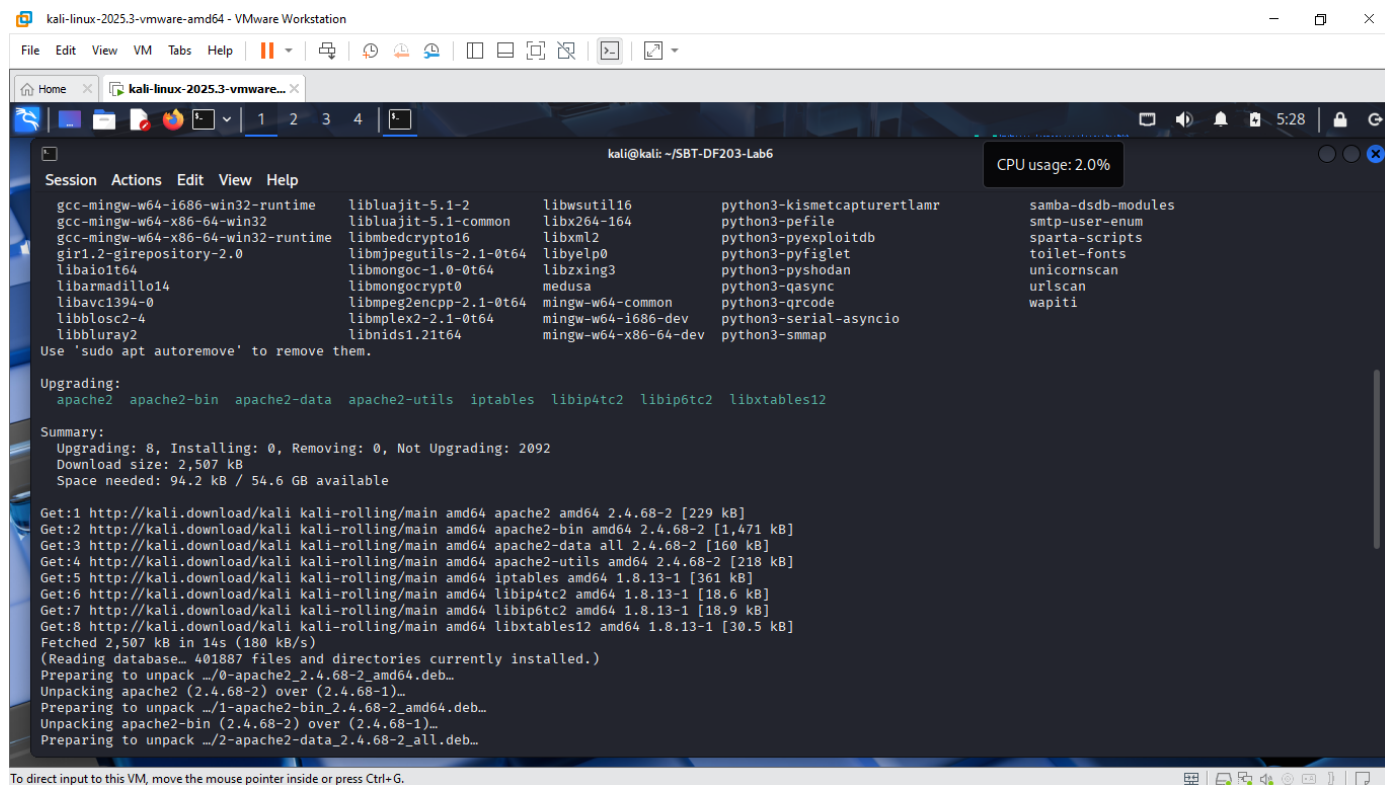
ANSWER:



```
kali@kali: ~/SBT-DF203-Lab6
(kali@kali)~$ sudo apt update
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.9 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [116 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [270 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [16.0 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [41.8 kB]
Fetched 77.0 MB in 1min 59s (649 kB/s)
2100 packages can be upgraded. Run 'apt list --upgradable' to see them.

(kali@kali)~$ sudo apt install -y apache2 curl iptables tshark wireshark
curl is already the newest version (8.21.0-2).
tshark is already the newest version (4.6.6-1).
wireshark is already the newest version (4.6.6-1).
The following packages were automatically installed and are no longer required:
amass-common libbson-1.0-0t64 libplacebo349 oracle-instantclient-basic python3-tld
binutils-mingw-w64-i686 libcrypt-dev libportmidi0 python3-bluepy python3-yaswfp
binutils-mingw-w64-x86-64 libgavl-1 libpython3.13-dev python3-click-plugins python3-zombie-imp
dnsmap libgdal37 libbrav1e0.7 python3-fs python3.13
dsniff libgirepository-1.0-1 libraw23t64 python3-git python3.13-dev
ettercap-common libimath-3-1-29t64 libsbm2-6 python3-gitdb python3.13-minimal
ettercap-graphical libjs-jquery-ui libtheoraec1 python3-gpg python3-kismetcapturebtgeiger python3.13-venv
figlet libjs-underscore libtheoraenc1 python3-kismetcapturefreaklabszigbee rsh-redone-client
finger libkmlbase1t64 libudfread0 python3-kismetcapturefreaklabszigbee samba-ad-dc
gcc-mingw-w64-base libkmlldom1t64 libwireshark18 python3-kismetcaptureertl433 samba-ad-provision
gcc-mingw-w64-i686-win32 libkmlengine1t64 libwiretap15 python3-kismetcaptureertladsb samba-dsdb-modules
gcc-mingw-w64-i686-win32-runtime liblua5.1-2 libwsutil16 python3-kismetcaptureertlamr
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



```
kali@kali: ~/SBT-DF203-Lab6
Session Actions Edit View Help
CPU usage: 2.0%

gcc-mingw-w64-i686-win32-runtime liblua5.1-2 libwsutil16 python3-kismetcaptureertlamr samba-dsdb-modules
gcc-mingw-w64-x86-64-win32 liblua5.1-2-common libx264-164 python3-pefile smtp-user-enum
gcc-mingw-w64-x86-64-win32-runtime libmbdedcrypto16 libxml2 python3-pyexploitdb sparta-scripts
gir1.2-girepository-2.0 libmjpegutils-2.1-0t64 libyelp0 python3-pyfiglet toilet-fonts
libaio1t64 libmongoc-1.0-0t64 libxzing3 python3-pyshodan urlscan
libarmadillo14 libmongocrypt0 medusa python3-qasync python3-qrcode unicornscan
libavc1394-0 libmpeg2encpp-2.1-0t64 mingw-w64-common python3-qrcode python3-serial-asyncio wrapti
libblosc2-4 libmpeg2-2.1-0t64 mingw-w64-i686-dev python3-qrcode python3-serial-asyncio wrapti
libbluray2 libnids1.21t64 mingw-w64-x86-64-dev python3-qrcode python3-serial-asyncio wrapti

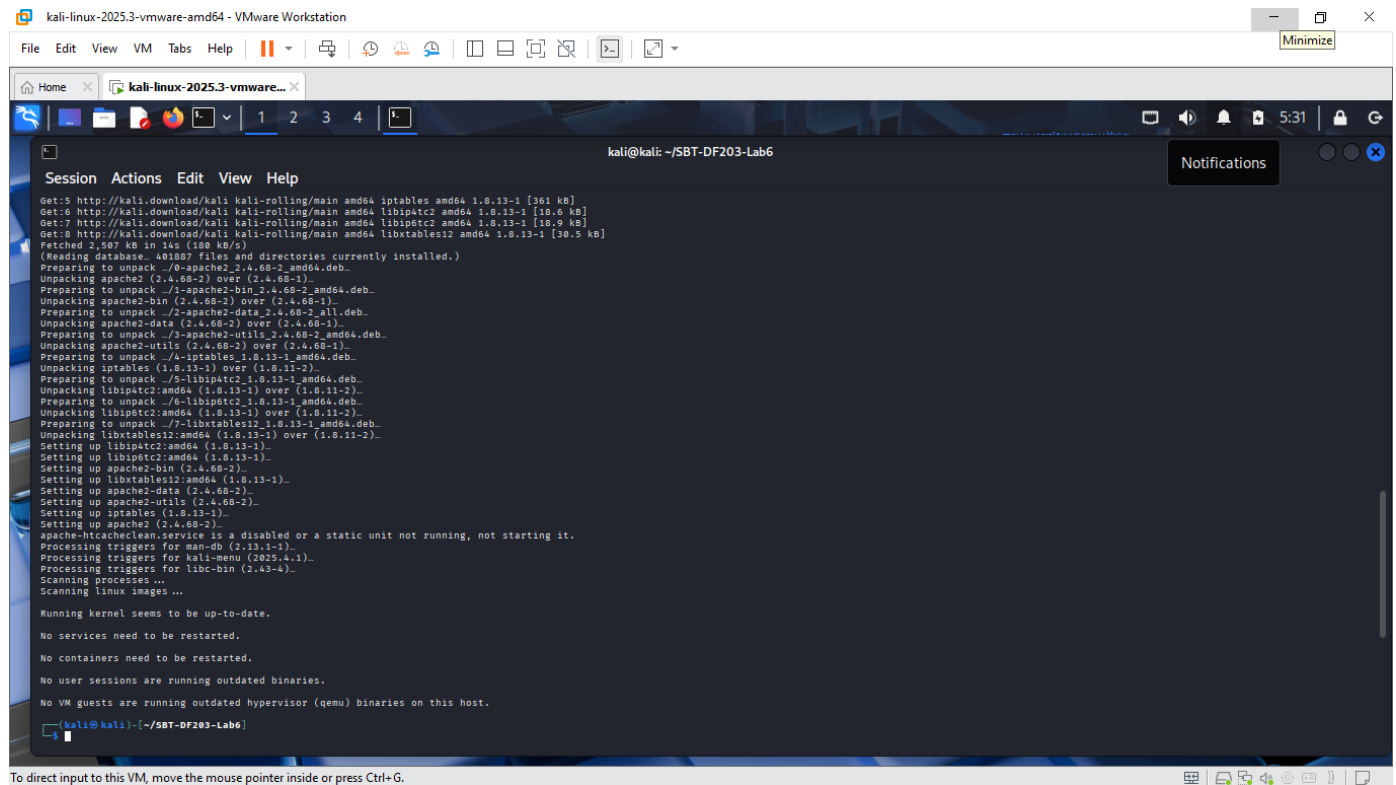
Use 'sudo apt autoremove' to remove them.

Upgrading:
apache2 apache2-bin apache2-data apache2-utils iptables libip4tc2 libip6tc2 libxtables12

Summary:
Upgrading: 8, Installing: 0, Removing: 0, Not Upgrading: 2092
Download size: 2,507 kB
Space needed: 94.2 kB / 54.6 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 apache2 amd64 2.4.68-2 [229 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 apache2-bin amd64 2.4.68-2 [1,471 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 apache2-data all 2.4.68-2 [160 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 apache2-utils amd64 2.4.68-2 [218 kB]
Get:5 http://kali.download/kali kali-rolling/main amd64 iptables amd64 1.8.13-1 [361 kB]
Get:6 http://kali.download/kali kali-rolling/main amd64 libip4tc2 amd64 1.8.13-1 [18.6 kB]
Get:7 http://kali.download/kali kali-rolling/main amd64 libip6tc2 amd64 1.8.13-1 [18.9 kB]
Get:8 http://kali.download/kali kali-rolling/main amd64 libxtables12 amd64 1.8.13-1 [30.5 kB]
Fetched 2,507 kB in 14s (180 kB/s)
(Reading database... 401887 files and directories currently installed.)
Preparing to unpack ./0-apache2_2.4.68-2_amd64.deb...
Unpacking apache2 (2.4.68-2) over (2.4.68-1)...
Preparing to unpack ./1-apache2-bin_2.4.68-2_amd64.deb...
Unpacking apache2-bin (2.4.68-2) over (2.4.68-1)...
Preparing to unpack ./2-apache2-data_2.4.68-2_all.deb...
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



kali-linux-2025.3-vmware-amd64 - VMware Workstation

File Edit View VM Tabs Help

Home kali-linux-2025.3-vmware... x

1 2 3 4

kali@kali: ~/SBT-DF203-Lab6

Session Actions Edit View Help

```
Get:5 http://kali.download/kali kali-rolling/main amd64 iptables amd64 1.8.13-1 [361 kB]
Get:6 http://kali.download/kali kali-rolling/main amd64 libipsetc2 amd64 1.8.13-1 [18.6 kB]
Get:7 http://kali.download/kali kali-rolling/main amd64 libipsetc2 amd64 1.8.13-1 [18.9 kB]
Get:8 http://kali.download/kali kali-rolling/main amd64 libxtables12 amd64 1.8.13-1 [30.5 kB]
Fetched 2,507 kB in 1s (180 kB/s)
(Reading database... 401887 files and directories currently installed.)
Preparing to unpack ./0-apache2_2.4.68-2_amd64.deb...
Unpacking apache2 (2.4.68-2) over (2.4.68-1)...
Preparing to unpack ./1-apache2-bin_2.4.68-2_amd64.deb...
Unpacking apache2-bin (2.4.68-2) over (2.4.68-1)...
Preparing to unpack ./2-apache2-data_2.4.68-2_all.deb...
Unpacking apache2-data (2.4.68-2) over (2.4.68-1)...
Preparing to unpack ./3-apache2-utils_2.4.68-2_amd64.deb...
Unpacking apache2-utils (2.4.68-2) over (2.4.68-1)...
Preparing to unpack ./4-iptables_1.8.13-1_amd64.deb...
Unpacking iptables (1.8.13-1) over (1.8.11-2)...
Preparing to unpack ./5-libipsetc2_1.8.13-1_amd64.deb...
Unpacking libipsetc2:amd64 (1.8.13-1) over (1.8.11-2)...
Preparing to unpack ./6-libipsetc2_1.8.13-1_amd64.deb...
Unpacking libipsetc2:amd64 (1.8.13-1) over (1.8.11-2)...
Preparing to unpack ./7-libxtables12_1.8.13-1_amd64.deb...
Unpacking libxtables12:amd64 (1.8.13-1) over (1.8.11-2)...
Setting up libipsetc2:amd64 (1.8.13-1)...
Setting up apache2-bin (2.4.68-2)...
Setting up libxtables12:amd64 (1.8.13-1)...
Setting up apache2-data (2.4.68-2)...
Setting up apache2-utils (2.4.68-2)...
Setting up iptables (1.8.13-1)...
Setting up apache2 (2.4.68-2)...
apache2-htcacheclean.service is a disabled or a static unit not running, not starting it.
Processing triggers for man-db (2.13.1-1)...
Processing triggers for libc-bin (2.43-4)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

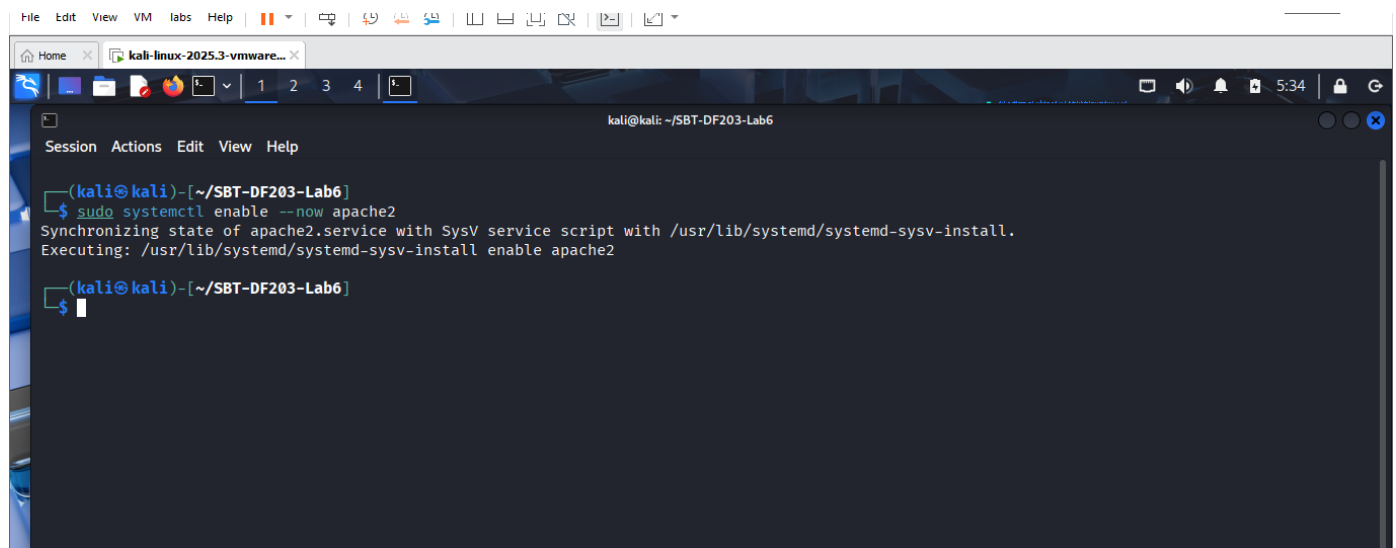
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

(kali@kali)~[/SBT-DF203-Lab6]
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



File Edit View VM Tabs Help

Home kali-linux-2025.3-vmware... x

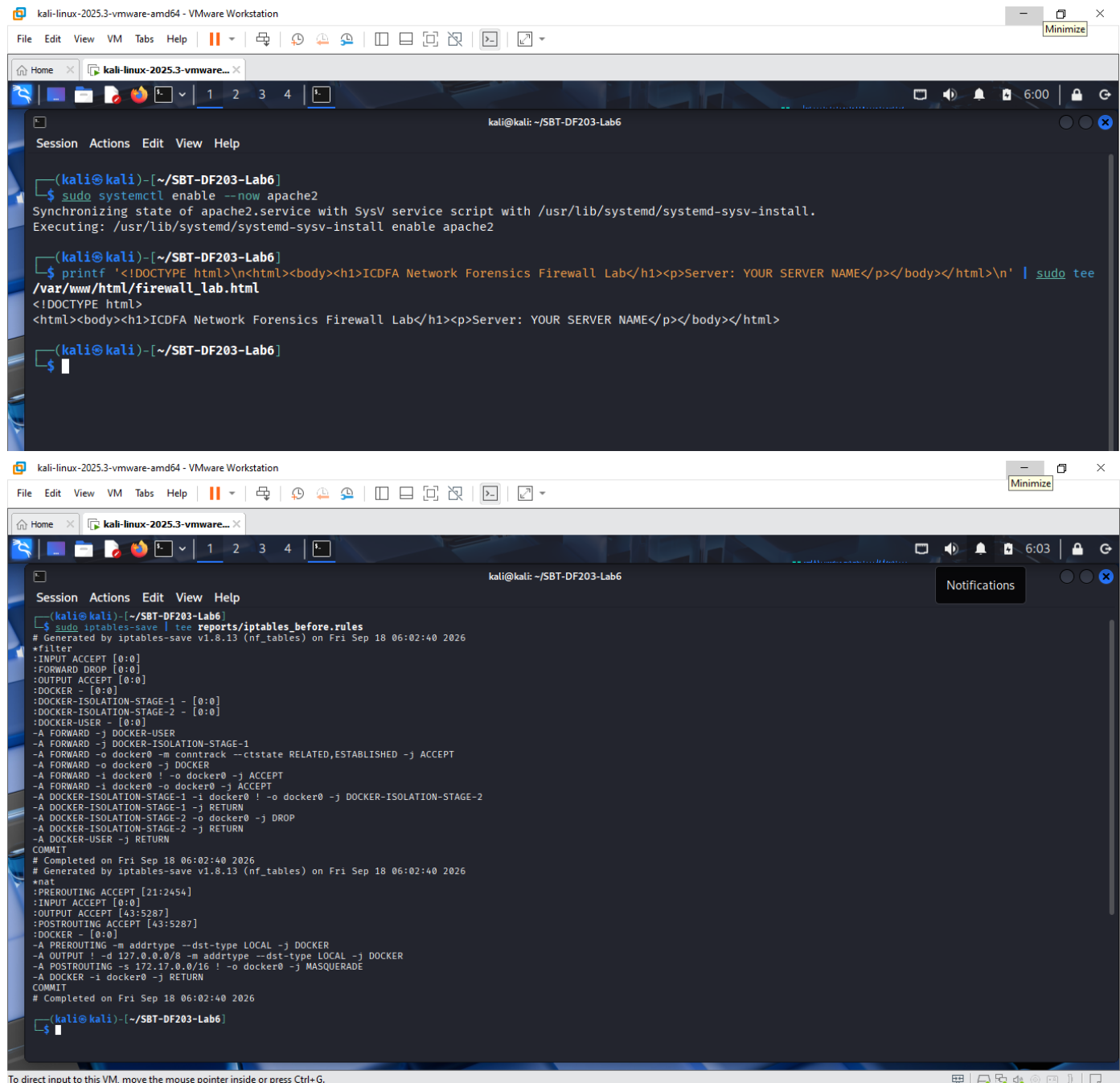
1 2 3 4

kali@kali: ~/SBT-DF203-Lab6

Session Actions Edit View Help

```
(kali@kali)~[/SBT-DF203-Lab6]
$ sudo systemctl enable --now apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2

(kali@kali)~[/SBT-DF203-Lab6]
$
```



```
kali@kali: ~/SBT-DF203-Lab6
$ sudo systemctl enable --now apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2

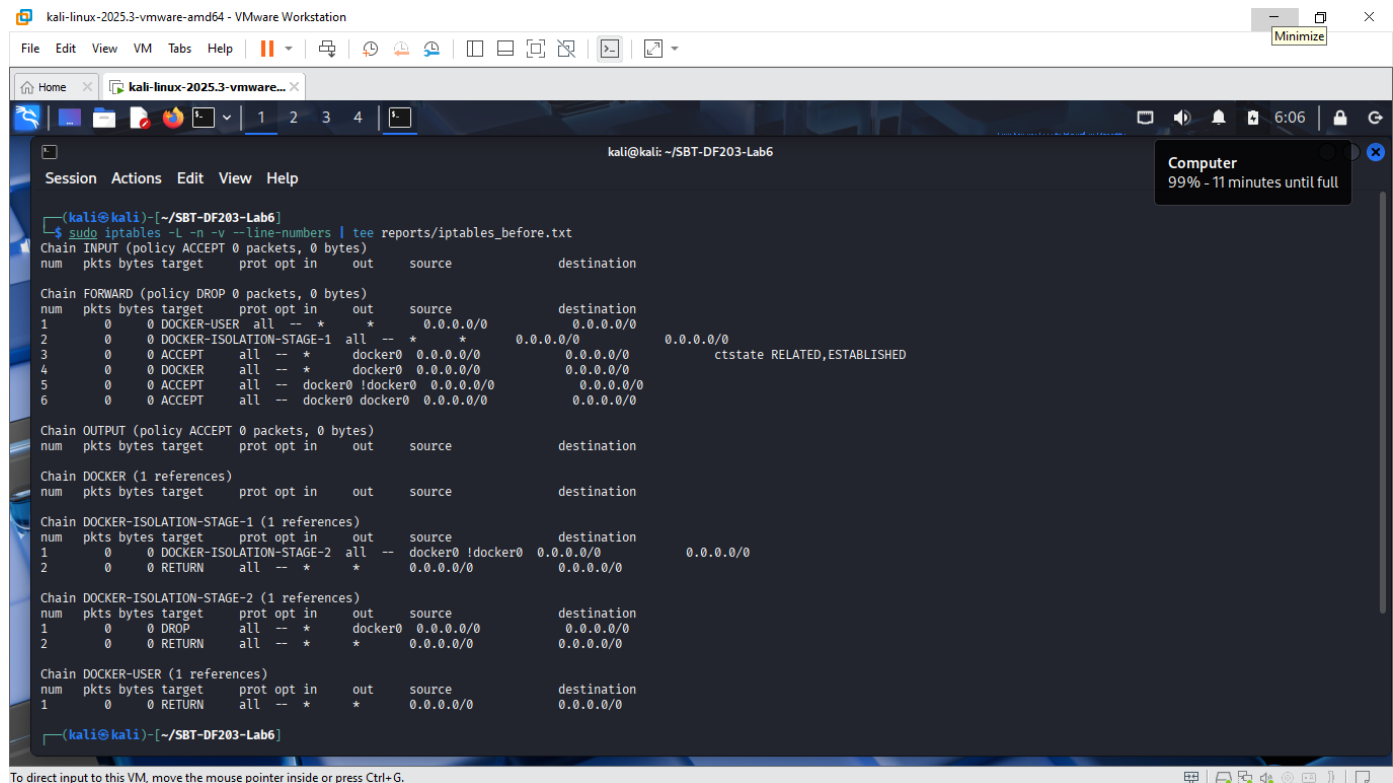
(kali@kali)~-[~/SBT-DF203-Lab6]
$ printf '<!DOCTYPE html>\n<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: YOUR SERVER NAME</p></body></html>\n' | sudo tee /var/www/html/firewall_lab.html
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: YOUR SERVER NAME</p></body></html>

(kali@kali)~-[~/SBT-DF203-Lab6]
$

kali@kali: ~/SBT-DF203-Lab6
$ sudo iptables-save | tee reports/iptables_before.rules
# Generated by iptables-save v1.8.13 (nf_tables) on Fri Sep 18 06:02:40 2026
*filter
:INPUT ACCEPT [0:0]
:FORWARD DROP [0:0]
:OUTPUT ACCEPT [0:0]
:DOCKER - [0:0]
:DOCKER-ISOLATION-STAGE-1 - [0:0]
:DOCKER-ISOLATION-STAGE-2 - [0:0]
:DOCKER-USER - [0:0]
-A FORWARD -j DOCKER-USER
-A FORWARD -j DOCKER-ISOLATION-STAGE-1
-A FORWARD -o docker0 -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A FORWARD -o docker0 -j DOCKER
-A FORWARD -i docker0 ! -o docker0 -j ACCEPT
-A FORWARD -i docker0 -o docker0 -j ACCEPT
-A DOCKER-ISOLATION-STAGE-1 -i docker0 ! -o docker0 -j DOCKER-ISOLATION-STAGE-2
-A DOCKER-ISOLATION-STAGE-1 -j RETURN
-A DOCKER-ISOLATION-STAGE-2 -o docker0 -j DROP
-A DOCKER-ISOLATION-STAGE-2 -j RETURN
-A DOCKER-USER -j RETURN
COMMIT
# Completed on Fri Sep 18 06:02:40 2026
# Generated by iptables-save v1.8.13 (nf_tables) on Fri Sep 18 06:02:40 2026
*nat
:PREROUTING ACCEPT [21:2454]
:INPUT ACCEPT [0:0]
:OUTPUT ACCEPT [43:5287]
:POSTROUTING ACCEPT [43:5287]
:DOCKER - [0:0]
-A PREROUTING -m addrtype --dst-type LOCAL -j DOCKER
-A OUTPUT ! -d 127.0.0.0/8 -m addrtype --dst-type LOCAL -j DOCKER
-A POSTROUTING -s 172.17.0.0/16 ! -o docker0 -j MASQUERADE
-A DOCKER -i docker0 -j RETURN
COMMIT
# Completed on Fri Sep 18 06:02:40 2026

(kali@kali)~-[~/SBT-DF203-Lab6]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



```
kali@kali: ~/SBT-DF203-Lab6
$ sudo iptables -L -n -v --line-numbers | tee reports/iptables_before.txt
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                 destination
1     0    0 DOCKER-USER all -- *      *       0.0.0.0/0              0.0.0.0/0
2     0    0 DOCKER-ISOLATION-STAGE-1 all -- *      *       0.0.0.0/0              0.0.0.0/0
3     0    0 ACCEPT    all -- *      *       docker0 0.0.0.0/0              0.0.0.0/0
4     0    0 DOCKER    all -- *      *       docker0 0.0.0.0/0              0.0.0.0/0
5     0    0 ACCEPT    all -- docker0 !docker0 0.0.0.0/0              0.0.0.0/0
6     0    0 ACCEPT    all -- docker0 docker0   0.0.0.0/0              0.0.0.0/0

Chain FORWARD (policy DROP 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                 destination
1     0    0 DOCKER-USER all -- *      *       0.0.0.0/0              0.0.0.0/0
2     0    0 DOCKER-ISOLATION-STAGE-1 all -- *      *       0.0.0.0/0              0.0.0.0/0
3     0    0 ACCEPT    all -- *      *       docker0 0.0.0.0/0              0.0.0.0/0
4     0    0 DOCKER    all -- *      *       docker0 0.0.0.0/0              0.0.0.0/0
5     0    0 ACCEPT    all -- docker0 !docker0 0.0.0.0/0              0.0.0.0/0
6     0    0 ACCEPT    all -- docker0 docker0   0.0.0.0/0              0.0.0.0/0

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                 destination

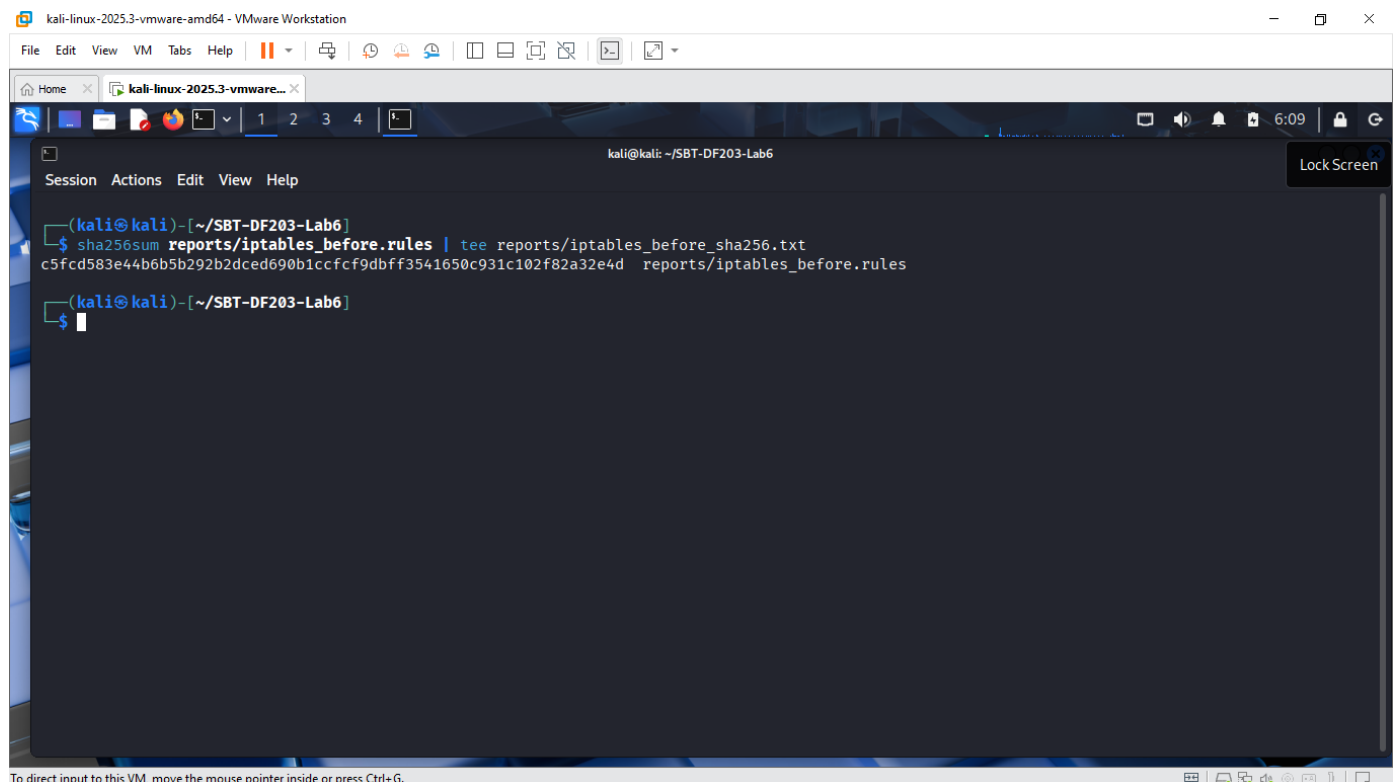
Chain DOCKER (1 references)
num  pkts bytes target    prot opt in     out     source                 destination

Chain DOCKER-ISOLATION-STAGE-1 (1 references)
num  pkts bytes target    prot opt in     out     source                 destination
1     0    0 DOCKER-ISOLATION-STAGE-2 all -- *      *       docker0 !docker0 0.0.0.0/0              0.0.0.0/0
2     0    0 RETURN    all -- *      *       0.0.0.0/0              0.0.0.0/0

Chain DOCKER-ISOLATION-STAGE-2 (1 references)
num  pkts bytes target    prot opt in     out     source                 destination
1     0    0 DROP     all -- *      *       docker0 0.0.0.0/0              0.0.0.0/0
2     0    0 RETURN    all -- *      *       0.0.0.0/0              0.0.0.0/0

Chain DOCKER-USER (1 references)
num  pkts bytes target    prot opt in     out     source                 destination
1     0    0 RETURN    all -- *      *       0.0.0.0/0              0.0.0.0/0
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



```
kali@kali: ~/SBT-DF203-Lab6
$ sha256sum reports/iptables_before.rules | tee reports/iptables_before_sha256.txt
c5fcd583e44b6b5b292b2dced690b1ccfcf9dbff3541650c931c102f82a32e4d reports/iptables_before.rules
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab6-ADOYI NATHANIEL JOHNSON
Trainee name	ADOYI NATHANIEL JOHNSON
Date and time started	18/09/26 09:10AM
Evidence file name(s)	SBT-DF203_LAB6_FIREWALL_TRAFFIC_CONTROL_AND_FORENSIC_VERIFICATION
Source or generation method	KALI COMMAND LINE
Original SHA-256	sha256sum reports/iptables_before.rules tee reports/iptables_before_sha256.txt
Working-copy SHA-256	c5fcd583e44b6b5b292b2dced690b1ccfcf9dbff3541650c931c102f82a32e4d reports/iptables_before.rules
Analysis workstation/VM	KALI@KALI
Notes on any changes	NIL

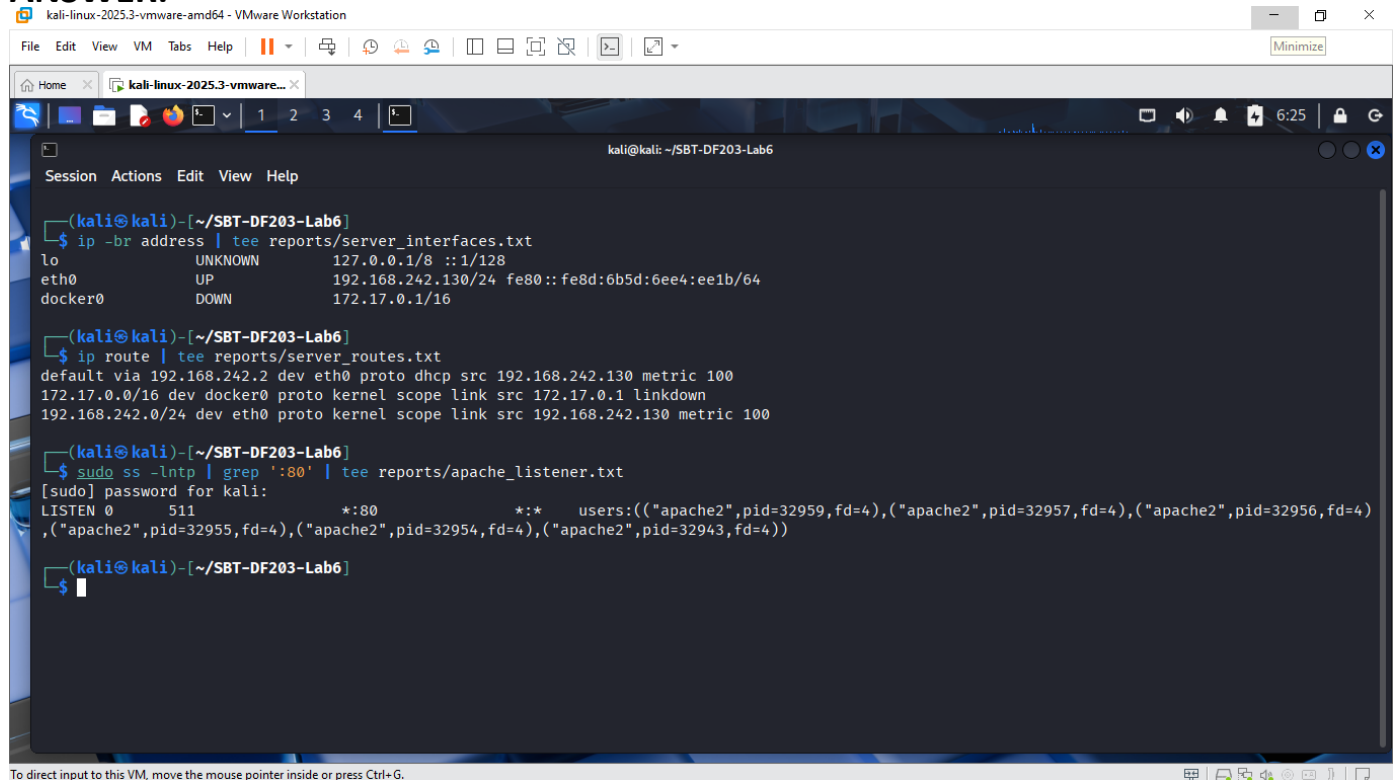
8. Part A - Record the Lab Network and Baseline Access

```
# On the server
ip -br address | tee reports/server_interfaces.txt
ip route | tee reports/server_routes.txt
sudo ss -lntp | grep ':80' | tee reports/apache_listener.txt

# On the client, replace SERVER_IP
curl -v --connect-timeout 5 http://SERVER_IP/firewall_lab.html 2>&1 | tee baseline_curl.txt
```

Record the server IP, allowed client IP, blocked client IP, interface and default gateway. Take a snapshot before changing the firewall.

ANSWER:

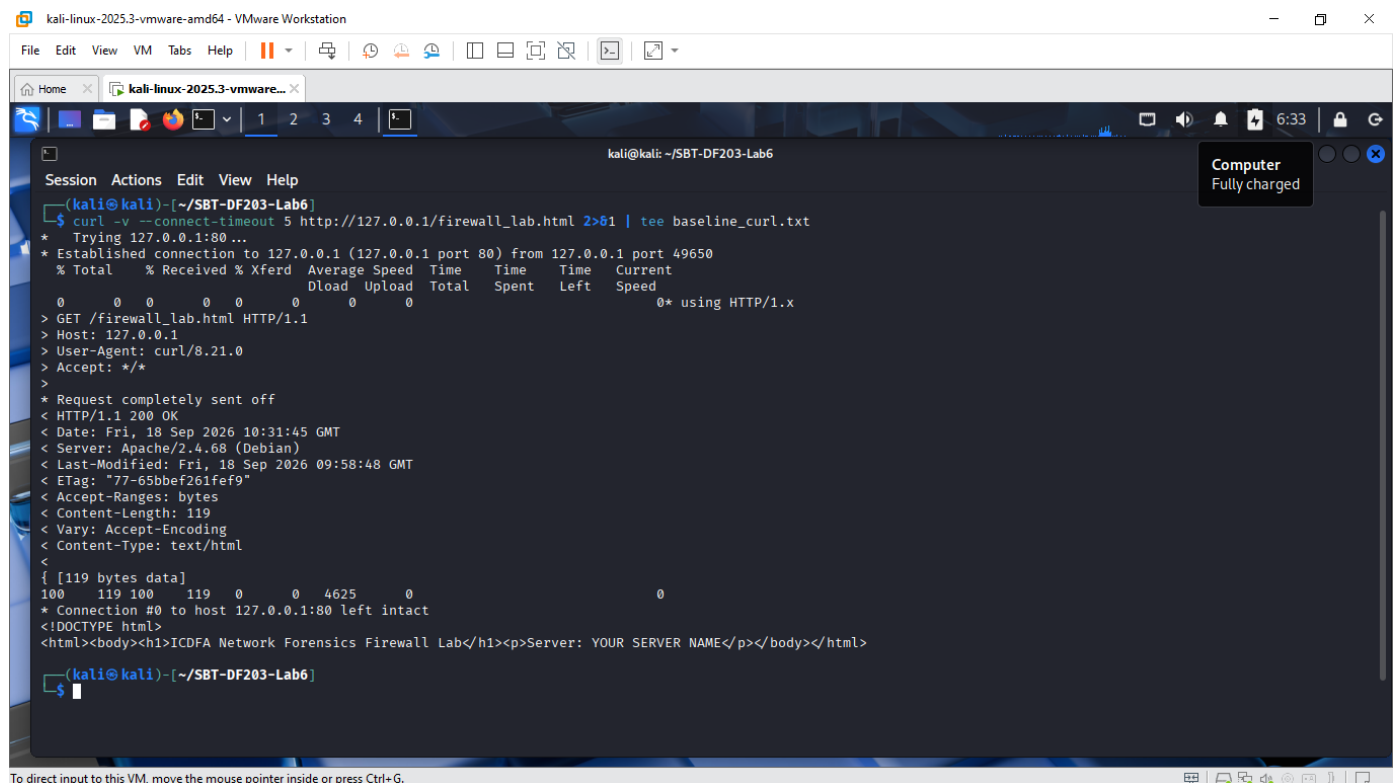


```
kali@kali: ~/SBT-DF203-Lab6
$ ip -br address | tee reports/server_interfaces.txt
lo                UNKNOWN      127.0.0.1/8 ::1/128
eth0              UP          192.168.242.130/24 fe80::fe8d:6b5d:6ee4:ee1b/64
docker0          DOWN        172.17.0.1/16

(kali@kali)~/SBT-DF203-Lab6
$ ip route | tee reports/server_routes.txt
default via 192.168.242.2 dev eth0 proto dhcp src 192.168.242.130 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.242.0/24 dev eth0 proto kernel scope link src 192.168.242.130 metric 100

(kali@kali)~/SBT-DF203-Lab6
$ sudo ss -lntp | grep ':80' | tee reports/apache_listener.txt
[sudo] password for kali:
LISTEN 0      511          *:80          *:~ users:(("apache2",pid=32959,fd=4),("apache2",pid=32957,fd=4),("apache2",pid=32956,fd=4),("apache2",pid=32955,fd=4),("apache2",pid=32954,fd=4),("apache2",pid=32943,fd=4))

(kali@kali)~/SBT-DF203-Lab6
$
```



```
kali@kali: ~/SBT-DF203-Lab6
$ curl -v --connect-timeout 5 http://127.0.0.1/firewall_lab.html 2>&1 | tee baseline_curl.txt
* Trying 127.0.0.1:80...
* Established connection to 127.0.0.1 (127.0.0.1 port 80) from 127.0.0.1 port 49650
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload  Total   Spent    Left     Speed
0   0    0    0    0     0      0      0  0* using HTTP/1.x
> GET /firewall_lab.html HTTP/1.1
> Host: 127.0.0.1
> User-Agent: curl/8.21.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Fri, 18 Sep 2026 10:31:45 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Fri, 18 Sep 2026 09:58:48 GMT
< ETag: "77-65bbef261fef9"
< Accept-Ranges: bytes
< Content-Length: 119
< Vary: Accept-Encoding
< Content-Type: text/html
<
{ [119 bytes data]
100  119 100   119    0     0  4625    0      0
* Connection #0 to host 127.0.0.1:80 left intact
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: YOUR SERVER NAME</p></body></html>

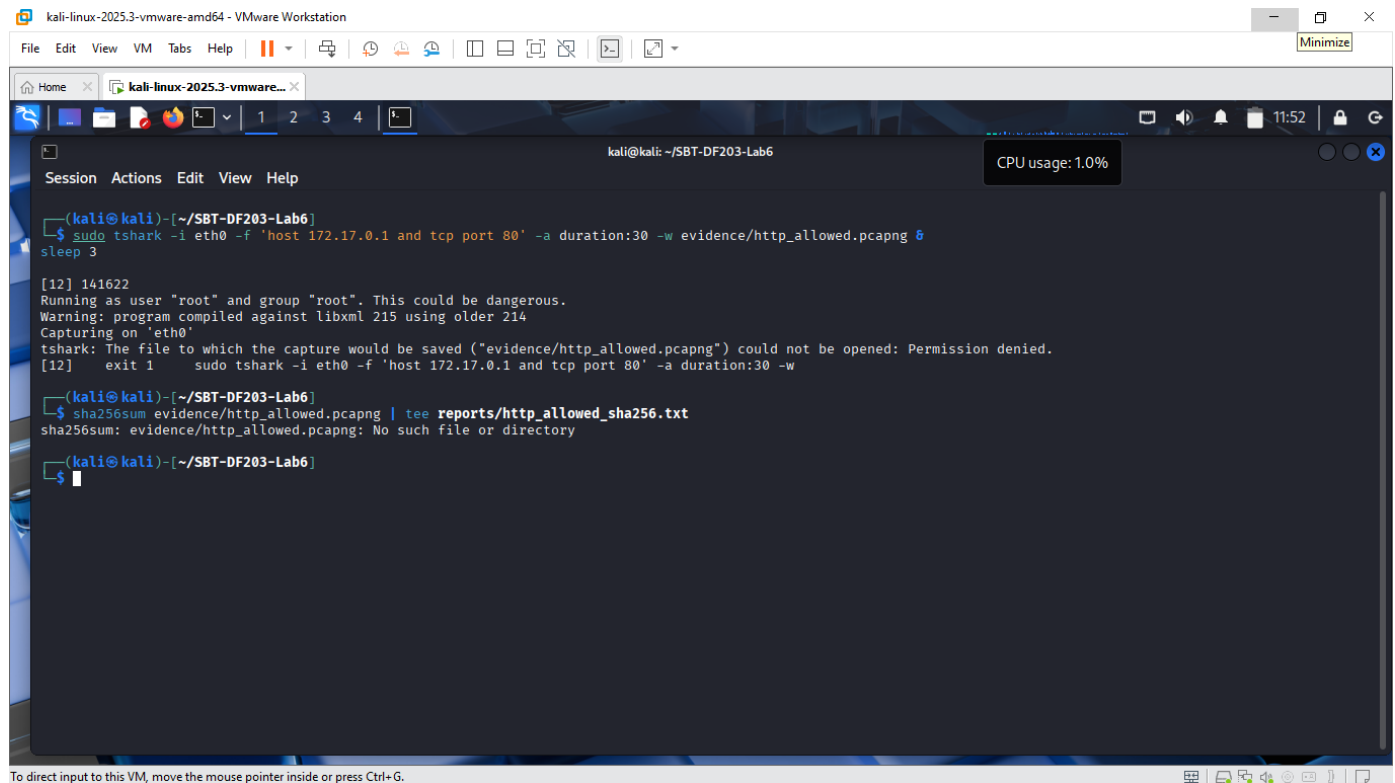
(kali@kali)~/SBT-DF203-Lab6
$
```

Computer Fully charged

9. Part B - Capture the Allowed HTTP Baseline

```
# On server, replace IFACE and CLIENT_IP
sudo tshark -i IFACE -f 'host CLIENT_IP and tcp port 80' -a duration:30 -w evidence/http_allowed.pcapng &
sleep 3
# Run curl from client during this capture
wait
sha256sum evidence/http_allowed.pcapng | tee reports/http_allowed_sha256.txt
```

ANSWER:

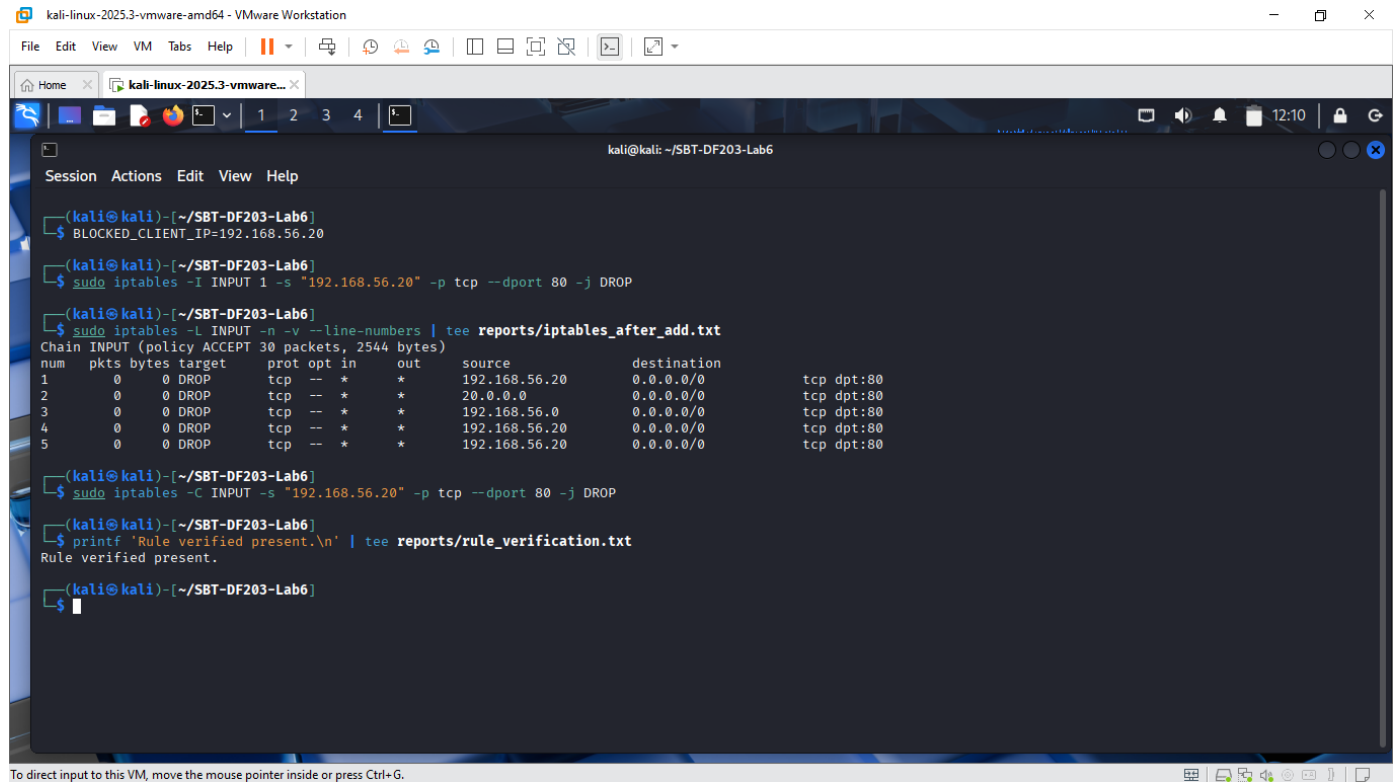


10. Part C - Apply and Verify the Blocking Rule

Use the exact source IP of the assigned blocked client. Insert the rule near the top of INPUT so it is evaluated before a broad ACCEPT rule.

```
BLOCKED_CLIENT_IP=192.168.56.20
sudo iptables -I INPUT 1 -s "$BLOCKED_CLIENT_IP" -p tcp --dport 80 -j DROP
sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_after_add.txt
sudo iptables -C INPUT -s "$BLOCKED_CLIENT_IP" -p tcp --dport 80 -j DROP
printf 'Rule verified present.\n' | tee reports/rule_verification.txt
```

ANSWER:



```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali@kali: ~/SBT-DF203-Lab6

(kali@kali)-[~/SBT-DF203-Lab6]
$ BLOCKED_CLIENT_IP=192.168.56.20

(kali@kali)-[~/SBT-DF203-Lab6]
$ sudo iptables -I INPUT 1 -s "192.168.56.20" -p tcp --dport 80 -j DROP

(kali@kali)-[~/SBT-DF203-Lab6]
$ sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_after_add.txt
Chain INPUT (policy ACCEPT 30 packets, 2544 bytes)
num  pkts bytes target    prot opt in     out     source               destination           tcp dpt:80
1      0      0 DROP      tcp  --  *      *       192.168.56.20        0.0.0.0/0             tcp dpt:80
2      0      0 DROP      tcp  --  *      *       20.0.0.0             0.0.0.0/0             tcp dpt:80
3      0      0 DROP      tcp  --  *      *       192.168.56.0         0.0.0.0/0             tcp dpt:80
4      0      0 DROP      tcp  --  *      *       192.168.56.20        0.0.0.0/0             tcp dpt:80
5      0      0 DROP      tcp  --  *      *       192.168.56.20        0.0.0.0/0             tcp dpt:80

(kali@kali)-[~/SBT-DF203-Lab6]
$ sudo iptables -C INPUT -s "192.168.56.20" -p tcp --dport 80 -j DROP

(kali@kali)-[~/SBT-DF203-Lab6]
$ printf 'Rule verified present.\n' | tee reports/rule_verification.txt
Rule verified present.

(kali@kali)-[~/SBT-DF203-Lab6]
$

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

11. Part D - Capture Blocked Traffic and Rule Counters

```

# On server
sudo tshark -i IFACE -f 'host 192.168.56.20 and tcp port 80' -a duration:35 -w evidence/http_blocked.pcapng &
sleep 3
# From blocked client during capture:
# curl -v --connect-timeout 10 http://SERVER_IP/firewall_lab.html
wait

sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_after_test.txt
sha256sum evidence/http_blocked.pcapng | tee reports/http_blocked_sha256.txt

```

ANSWER:

```

kali@kali: ~/SBT-DF203-Lab6
$ sudo tshark -i IFACE -f 'host 192.168.56.20 and tcp port 80' -a duration:35 -w evidence/http_blocked.pcapng &
sleep 3

[12] 152875
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'IFACE'
tshark: There is no device named "IFACE".
(No such device exists)
[12] exit 1 sudo tshark -i IFACE -f 'host 192.168.56.20 and tcp port 80' -a duration:35 -

kali@kali: ~/SBT-DF203-Lab6
$ curl -v --connect-timeout 10 http://SERVER_IP/firewall_lab.html
wait

* Resolving timed out after 10014 milliseconds
* closing connection #0
curl: (28) Resolving timed out after 10014 milliseconds

kali@kali: ~/SBT-DF203-Lab6
$ sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_after_test.txt
Chain INPUT (policy ACCEPT 57 packets, 4824 bytes)
num  pkts bytes target    prot opt in     out     source            destination
1      0      0 DROP      tcp  --  *      *        192.168.56.20      0.0.0.0/0          tcp dpt:80
2      0      0 DROP      tcp  --  *      *        20.0.0.0           0.0.0.0/0          tcp dpt:80
3      0      0 DROP      tcp  --  *      *        192.168.56.0       0.0.0.0/0          tcp dpt:80
4      0      0 DROP      tcp  --  *      *        192.168.56.20      0.0.0.0/0          tcp dpt:80
5      0      0 DROP      tcp  --  *      *        192.168.56.20      0.0.0.0/0          tcp dpt:80

```

```

kali@kali: ~/SBT-DF203-Lab6
$ sha256sum evidence/http_blocked.pcapng | tee reports/http_blocked_sha256.txt
sha256sum: evidence/http_blocked.pcapng: No such file or directory

kali@kali: ~/SBT-DF203-Lab6
$

```

12. Part E - Compare Allowed and Blocked Captures

```

for PCAP in evidence/http_allowed.pcapng evidence/http_blocked.pcapng; do
echo "==== $PCAP ====="
tshark -r "$PCAP" -Y 'tcp.flags.syn==1 || http.request || http.response || tcp.analysis.retransmission' \
-T fields -e frame.number -e frame.time_relative -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport \
-e tcp.flags -e tcp.analysis.retransmission -e http.request.uri -e http.response.code
done | tee reports/allowed_vs_blocked.tsv

```

Indicator	Allowed Capture	Blocked Capture
Client SYN visible?	YES client initiate TCP HANDSHAKE	YES client attempt SYN

Indicator	Allowed Capture	Blocked Capture
Server SYN-ACK visible?	YES, server responds	NO, Firewall drops packet
Handshake completed?	YES, SYN → SYN-ACK → ACK	NO, Handshake fails
HTTP GET visible?	YES, client sends request	NO, request never sent
HTTP response visible?	YES, server replies, up to 200 ok	NO, no response
Retransmissions/timeouts	YES, but normal	YES, repeated SYNs Timeout
iptables counter change	No drop counters increment	YES, DROP rule counters increase
curl result	Success page retrieved	Failure, connection timed out

ANSWER:

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
kali@kali: ~/SBT-DF203-Lab6
Session Actions Edit View Help
(kali@kali)-[~/SBT-DF203-Lab6]
$ echo "==== $PCAP ===="
(kali@kali)-[~/SBT-DF203-Lab6]
$ tshark -r "$PCAP" -Y 'tcp.flags.syn==1 || http.request || http.response || tcp.analysis.retransmission' \
  -T fields -e frame.number -e frame.time_relative -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport \
  -e tcp.flags -e tcp.analysis.retransmission -e http.request.uri -e http.response.code
Warning: program compiled against libxml 215 using older 214
tshark: The file "" doesn't exist.
(kali@kali)-[~/SBT-DF203-Lab6]
$ tee reports/allowed_vs_blocked.tsv
^C
(kali@kali)-[~/SBT-DF203-Lab6]
$

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

13. Part F - Remove the Rule and Restore Access

```

BLOCKED_CLIENT_IP=192.168.56.20
sudo iptables -D INPUT -s "$BLOCKED_CLIENT_IP" -p tcp --dport 80 -j DROP
sudo iptables -C INPUT -s "$BLOCKED_CLIENT_IP" -p tcp --dport 80 -j DROP || echo 'Rule successfully removed'
sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_restored.txt
# Confirm access from the previously blocked client
# curl -v --connect-timeout 5 http://SERVER_IP/firewall_lab.html

```

ANSWER:

```

kali@kali: ~/SBT-DF203-Lab6
$ BLOCKED_CLIENT_IP=192.168.56.20

(kali@kali)~/SBT-DF203-Lab6
$ sudo iptables -D INPUT -s "$BLOCKED_CLIENT_IP" -p tcp --dport 80 -j DROP
[sudo] password for kali:

(kali@kali)~/SBT-DF203-Lab6
$ sudo iptables -C INPUT -s "$BLOCKED_CLIENT_IP" -p tcp --dport 80 -j DROP || echo 'Rule successfully removed'

(kali@kali)~/SBT-DF203-Lab6
$ sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_restored.txt
Chain INPUT (policy ACCEPT) 194 packets, 16837 bytes
num  pkts bytes target    prot opt in     out     source               destination           tcp dpt:80
1      0      0 DROP      tcp  --  *      *       20.0.0.0             0.0.0.0/0             tcp dpt:80
2      0      0 DROP      tcp  --  *      *       192.168.56.0         0.0.0.0/0             tcp dpt:80
3      0      0 DROP      tcp  --  *      *       192.168.56.20        0.0.0.0/0             tcp dpt:80
4      0      0 DROP      tcp  --  *      *       192.168.56.20        0.0.0.0/0             tcp dpt:80
  
```

```

kali@kali: ~/SBT-DF203-Lab6
$ curl -v --connect-timeout 5 http://SERVER_IP/firewall_lab.html
* Resolving timed out after 5017 milliseconds
* closing connection #0
curl: (28) Resolving timed out after 5017 milliseconds

(kali@kali)~/SBT-DF203-Lab6
$
  
```

14. Part G - Optional NFQUEUE Observation**Instructor-Led Extension**

The slide introduces NFQUEUE and Python packet callbacks. Complete this only in the prepared lab VM because package names and Python bindings vary. The core assessed task is iptables rule evidence and PCAP verification.

- Explain that NFQUEUE passes matching packets to a userspace program.

ANSWER: is a special iptables target that sends matching packets from the kernel into userspace.

- Identify where a callback can accept, modify or drop the packet.
- Document the rule, queue number, process and cleanup command.
- Never leave an NFQUEUE rule active without a listening process because traffic may stall.

15. Forensic Interpretation Questions

1. Why does DROP commonly cause SYN retransmissions and a timeout?
2. How would a REJECT rule differ in the packet capture and client output?
3. Why are firewall counters valuable corroborating evidence?
4. What evidence would show that the web server was down rather than firewall-blocked?
5. What is the risk of deleting a rule by line number after other rules have changed?

Required Screenshots Checklist

- ☐ Original ruleset and hash.
- ☐ Network addresses and Apache listener.
- ☐ Successful baseline curl.
- ☐ Allowed HTTP capture handshake/request/response.
- ☐ Inserted DROP rule and verification.
- ☐ Blocked curl timeout.
- ☐ Blocked capture showing retransmissions/no response.
- ☐ Rule packet/byte counters after test.
- ☐ Allowed-versus-blocked comparison.
- ☐ Rule removal and restored access.

Student Report Template

6. Executive summary.
7. Lab network and scope.
8. Original firewall state and evidence hashes.
9. Allowed baseline analysis.
10. Rule implementation and verification.
11. Blocked traffic analysis and counter correlation.
12. DROP versus REJECT discussion.
13. Cleanup and restored service.
14. Conclusions and appendix.

Submission Requirements

- One PDF report named SBT-DF203-Lab6_RegNo_FullName.pdf.
- The original or generated PCAP/PCAPNG evidence file and its SHA-256 hash text file.
- A reports folder containing exported CSV/TXT command output and any recovered objects.
- Screenshots must be readable, numbered and referenced in the report narrative.
- Compress the report and evidence outputs into one ZIP named with your registration number and lab number.
- Do not submit passwords or decoded credentials in public discussion channels; include them only in the protected assessment submission where required.

Marking Rubric (100 Marks)

Assessment Area	Marks	What the Instructor Will Check
Safety and ruleset preservation	15	Original rules exported; narrow rule; clean restoration.
Baseline capture	15	Complete allowed HTTP session.
Firewall rule execution	20	Correct insertion, verification, counters and removal.
Blocked traffic forensics	25	Accurate packet/curl/counter correlation.

Assessment Area	Marks	What the Instructor Will Check
Interpretation	15	DROP, REJECT, service-down and evidence limitations.
Report quality	10	Professional screenshots and narrative.

Instructor Quick Answer Guide

Checkpoint	Expected Observation
INPUT chain	Traffic destined for the local host.
DROP behaviour	Packet silently discarded; client often retransmits then times out.
REJECT behaviour	Firewall sends an error/reset, producing faster failure.
Counter evidence	Packet/byte counters increase when the rule matches.
Allowed session	SYN, SYN-ACK, ACK, HTTP request and response.
Cleanup	Delete the exact rule specification and confirm access returns.

Command Reference Summary

Command	Purpose
iptables -L -n -v --line-numbers	List rules and counters.
iptables-save	Export the current ruleset.
iptables -I INPUT 1 ... -j DROP	Insert a narrow blocking rule.
iptables -C ...	Check whether a rule exists.
iptables -D ...	Delete an exact rule.
tcp.analysis.retransmission	Filter retransmitted TCP packets.

Academic Integrity and Authorization Statement

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.